

Signature Verification Product Architecture

What to use: Computer Vision (CV), Classical Machine Learning (ML), Deep Learning (DL), and the Verification Layer

Target: Real-time verification from a scanned/photographed signature image, supporting two modes: (1) at least 3 genuine + forged references, and (2) at least 3 genuine references with no forged samples.

Core principle: Treat the system as a *verification* system, not merely a binary image classifier. A questioned signature is compared with a claimed writer's genuine reference profile. A single feature never proves authenticity.

1. What each technology is doing

Layer	Role	Where used
Computer Vision (CV)	Reads the static signature image and extracts measurable visual/structural information: segmentation, thresholding, normalization, contours, skeletons, geometry, projections, HOG/LBP/GLCM, moments, etc.	Input processing + feature extraction
Classical Machine Learning (ML)	Learns a decision or distance boundary from extracted features. Examples: SVM, Random Forest, XGBoost, Logistic Regression, One-Class SVM, Mahalanobis/prototype methods.	Verification/decision model
Deep Learning (DL)	Learns a high-level signature representation directly from images. CNNs and Siamese/contrastive/metric-learning networks are suitable for signature embeddings.	Representation learning + similarity
Verification layer	Compares a query against multiple genuine references, models natural intra-writer variation, calibrates thresholds and returns a score/result.	Identity-specific decision
Evaluation/statistics	Measures FAR, FRR, EER, ROC/DET and performance on random/skilled forgeries.	Scientific validation

2. Recommended real-time pipeline

Camera / uploaded scan

→ Image-quality check (resolution, blur, contrast, background, crop)

→ Signature segmentation

→ Grayscale + binarization/foreground extraction

→ Crop to signature bounding box

→ Scale/position normalization

→ CV feature extraction

→ DL embedding extraction

→ Compare query with the person's genuine reference profile

→ Verification score + quality flag

→ Genuine / not consistent with reference

3. CV features you should actually implement first

CV family	Features
Global geometry	width, height, aspect ratio/calibre, foreground area, bounding-box area, ink density, perimeter, centroid X/Y, extent
Spatial/graphometric	baseline angle/deviation, slant, slant variance, component spacing, negative-space distribution, component size/position statistics
Structural	connected components, skeleton length, endpoints, junctions, branches, loops, crossings, connectivity, turning-point proxies
Contour/shape	contour length, curvature statistics, circularity, solidity, convex-hull ratio, eccentricity, Hu moments
Directional	local orientation, orientation histogram, HOG, directional distribution
Projection/grid	horizontal projection, vertical projection, 4×4 or 8×8 regional ink density/statistics

CV family	Features
Texture (secondary)	LBP, GLCM contrast/correlation/energy/homogeneity/entropy
Transforms (optional)	Fourier, wavelet, DCT or Radon descriptors

4. What is forensic/graphometric vs computer-vision

Forensic/graphometric characteristics: size, proportions, spacing, slant, baseline/alignment, character/component shape and construction, initial and terminal strokes, connections, fluency/line quality, natural variation, and structural stroke characteristics. **Computer-vision descriptors:** contours, skeleton graphs, moments, HOG, LBP, GLCM, projections, grids and transform coefficients. The second group is a computational representation of the image; it should not be presented as a forensic standard parameter list.

5. Deep Learning: what I recommend

Use DL primarily to learn a **writer-independent signature embedding**, not to train a separate CNN from scratch for each person. A CNN/vision backbone maps a signature image to an embedding vector. A Siamese or contrastive/metric-learning setup can learn that genuine signatures from the same writer should be close while different writers/forgeries should be farther apart.

Important for your 3-sample requirement: 3 genuine signatures are too few to train a new deep network from scratch. The network should be pretrained/generalized using a sufficiently large multi-writer dataset; the user's 3+ genuine signatures should then create a person-specific reference profile.

6. Model A — genuine + forged references

Goal: use genuine and known-forgery examples to learn a writer-specific boundary or calibrate a verifier.

- Input: ≥ 3 genuine + available forged signatures.
- CV path: extract the core handcrafted feature vector.
- DL path: extract the pretrained signature embedding.
- Combine or evaluate the two representations separately first.
- Candidate models: SVM/XGBoost/Random Forest for handcrafted features; metric-learning/Siamese similarity for embeddings.
- Compare query-to-reference distances, not only query-to-image classification.
- Evaluate separately on genuine, random forgery and skilled forgery where available.

7. Model B — genuine-only references

Goal: no forged samples are available, so this is a one-class/anomaly-style verification problem.

- Input: ≥ 3 genuine signatures.
- Extract CV features and/or DL embeddings.
- Create a genuine profile/prototype from the reference signatures.
- Estimate natural genuine-to-genuine variation.
- Compare the query against the profile and its expected variation.
- Candidate methods: prototype/cosine or Euclidean distance, Mahalanobis distance, One-Class SVM, or an embedding-based calibrated threshold.
- Output should preferably be 'consistent with reference' vs 'not consistent with reference', with an uncertainty/quality flag.

8. How the 3 genuine signatures should be used

Do not simply train a binary classifier with three positive images. Compute reference representations E1, E2 and E3. Measure $E1 \leftrightarrow E2$, $E1 \leftrightarrow E3$ and $E2 \leftrightarrow E3$ to estimate natural variation. Then compare the query embedding EQ to all three references and/or a robust profile. The decision threshold should be validated empirically rather than chosen as an arbitrary value such

as 80%.

If possible, support 5 or 10+ genuine references. Three is a minimum operating condition, not an ideal statistical sample.

9. Features NOT directly available from an ordinary scan/photo

Do **not** claim direct measurement of actual pen pressure, true writing velocity, acceleration, exact signing duration, exact pen-up duration, or pen inclination from a static image. These are online/dynamic signals that require appropriate digital capture. Static images can provide visual proxies, but those proxies are not the original physical measurements.

10. Real-time performance strategy

- Keep preprocessing lightweight: crop, grayscale/threshold, normalize.
- Precompute each user's reference embeddings and handcrafted feature statistics when they enroll.
- At verification time, process only the new signature.
- Use a small/efficient inference model and cache reference representations.
- Do not recompute expensive transforms for every reference if their features are already stored.
- Run the image-quality gate first so poor images are rejected before expensive inference.
- Return a result only after both quality and similarity checks pass.

11. Recommended final architecture

Enrollment: 3+ genuine signatures → preprocessing → CV features + DL embeddings → store genuine profile + intra-writer variation.

Verification: query image → quality gate → same preprocessing → CV features + DL embedding → compare with stored profile → calibrated decision.

For Model A, known forgeries can be used for additional supervised calibration. For Model B, use one-class/profile-based verification because no negative examples are available.

12. Recommended implementation roadmap

Phase	Focus	Work
Phase 1	CV baseline	Preprocessing + 30–50 core features + distance-based verification.
Phase 2	Classical ML	SVM/XGBoost/Random Forest on handcrafted features; compare against simple distance methods.
Phase 3	Genuine-only	Prototype/Mahalanobis/One-Class SVM and personalized thresholds.
Phase 4	DL	Train/fine-tune a general signature embedding model using multi-writer data.
Phase 5	Hybrid	Compare CV score + DL embedding score; calibrate a combined verifier.
Phase 6	Real-time	Cache enrollment representations; optimize preprocessing/inference.
Phase 7	Validation	FAR, FRR, EER, ROC/DET; random vs skilled forgeries; writer-disjoint test protocol.

13. Research basis

- ANSI/ASB Standard 070-22, Standard for Examination of Handwritten Items (2022): <https://www.aafs.org/asb-standard/standard-examination-handwritten-items>
- Hafemann, Sabourin & Oliveira, Offline Handwritten Signature Verification – Literature Review: <https://arxiv.org/abs/1507.07909>
- Impedovo & Pirlo, Automatic Signature Verification: The State of the Art, IEEE TSMC-C (2008): <https://doi.org/10.1109/TSMCC.2008.923866>
- Hafemann, Sabourin & Oliveira, Learning features for offline handwritten signature verification using deep convolutional neural networks, Pattern Recognition (2017): <https://doi.org/10.1016/j.patcog.2017.05.012>
- Ebrahim et al., Features Selection for Offline Handwritten Signature Verification: State of the Art (2018): <https://doi.org/10.1504/IJCVR.2018.095590>

14. Final recommendation

Use CV for measurement, DL for representation, ML/statistical methods for decision-making, and a dedicated verification layer for the person's reference profile. Do not make the product a simple 'real/fake image classifier'. The strongest design for your constraints is a writer-independent representation plus writer-specific verification.